

MUHAMMED SALIM K T

Senior Python Developer · AI/ML Engineer (LLM, RAG & Speech)

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Target: UAE (Dubai / Abu Dhabi), on-site or hybrid **Nationality:** Indian **Relocation:** Ready to relocate to the UAE
Experience: 4 years 4 months (Python, production)
Languages: English (fluent), Arabic (proficient: reading, writing, comprehension), Malayalam, Hindi, Urdu

PROFESSIONAL SUMMARY

Python engineer with 4+ years building and running production backend and AI systems on **FastAPI, Django, Celery, Redis and PostgreSQL**. I built the speech-AI service behind a US telephony provider's voicemail and call-recording transcription, wrote the multi-agent tool-calling engine that replaced CrewAI in a live enterprise platform (a 12B model now does work that previously needed 32B), and designed RAG over **10M+ embeddings serving 10,000+ queries/day** at sub-2-second latency. Experienced in on-premise, air-gapped delivery to regulated clients and in Kubernetes, Docker and GCP; proficient in Arabic, which helps with Arabic LLM and search work for Gulf products.

CORE SKILLS

Python & backend	Python 3 (4+ yrs), FastAPI, Django, Flask, Celery (priority queues, worker pools), asyncio, Pydantic, REST, WebSockets, SSE, RabbitMQ, Kafka, Node.js, Java Spring Boot
AI / LLM	RAG design, hybrid search and re-ranking, multi-agent orchestration, schema-validated tool calling, MCP, LangChain, LlamaIndex, CrewAI, prompt engineering, LLM evals and regression gates
ML & models	PyTorch, Hugging Face Transformers, Sentence-Transformers, fine-tuning (Unsloth, LoRA / QLoRA, PEFT; BERT, T5, GPT-family), quantization, scikit-learn, pandas, NumPy
Model serving	vLLM, Ollama, OpenAI-compatible APIs, load-balanced inference tiers with failover, batching and caching, on-prem / air-gapped deployment
Speech & vision	Whisper, NVIDIA Canary, speaker diarization (Silero VAD, SpeechBrain ECAPA-TDNN), Tacotron2, PaddleOCR-VL, Qwen-VL, LayoutLMv3, Docling, RapidOCR, MediaPipe
Data & vector DBs	PostgreSQL + pgvector, Cassandra (vector), Qdrant, FAISS, Pinecone, MongoDB, Redis, MySQL, DuckDB, SQLAlchemy, ETL
Cloud & DevOps	Kubernetes (Kustomize, GPU/DGX nodes, cert-manager), Docker, GCP (Cloud Run, Vertex AI), GitLab CI, GitHub Actions, Jenkins, Prometheus, Grafana, Git, Agile / Scrum
Frontend & mobile	TypeScript, React, React Native (custom turbo modules in Kotlin / Objective-C)

PROFESSIONAL EXPERIENCE

AI/ML Engineer & Full-Stack Developer (Python) · Netstratum Technologies Pvt Ltd, Kochi, India

June 2022 – Present

Python lead for BlueMesh, the company's on-premise enterprise AI platform, plus speech and real-time products for US enterprise clients.

SPEECH AI: ECHO (FLAGSHIP, IN PRODUCTION)

- Built **Echo** in Python, a production speech-AI service transcribing voicemails and call recordings for a US hosted-telephony provider. It runs **NVIDIA Canary** for accuracy-first voicemail and **Whisper on vLLM** for call recordings, with automatic engine failover, on **FastAPI + Cassandra**.
- Designed the throughput path: **Redis priority queues (0–99)** with live re-prioritisation and **Celery** workers on stateless HA nodes, with duplicate suppression and a CUDA-error watchdog.
- Engineered the audio pipeline: 45-minute chunking of recordings up to 10 hours with absolute timestamps, automatic language ID, speaker diarization (per-channel VAD for stereo; ECAPA-TDNN + clustering for mono), and Gemma LLM summaries on a separate queue so they never block transcription.

LLM, RAG & AGENTIC SYSTEMS

- Built and shipped **BlueMesh**, an on-premise AI chat platform in **Python / FastAPI**, serving **10,000+ queries/day** at **sub-2 s** response times with no customer data leaving the client's infrastructure.
- Designed its RAG pipeline end to end: Docling + RapidOCR ingestion, chunking, Sentence-Transformers embeddings, Cassandra vector schema and hybrid keyword + semantic retrieval over **10M+ embeddings**. Answer accuracy improved **45%** over the bare LLM baseline, and pgvector and FAISS search cut retrieval time **70%**.

- Reduced average inference latency to **1.8 s** with quantization, batching and caching, and held **99.7% uptime** with model failover and warm caching.
- Authored **bm-agent-engine**, a versioned Python package that replaced CrewAI / LangChain and now serves all agent traffic. Re-architecting the orchestration lets a **12B model do 32B-class work**, which moved each deployment from a GPU cluster to a single node.
- Built a tool registry exposing **73 toolsets** (GitLab, SharePoint / MS Graph, AWS, Azure, BigQuery, Jira, Snowflake, Kafka, Playwright). Added ReAct agents with tiered escalation, a FAST / SMALL / LARGE / THINKING model router over vLLM and Ollama, SSE streaming and Redis-backed admission control.
- Fine-tuned Hugging Face models (BERT, T5, GPT-family, plus Arabic LLMs with LoRA / QLoRA) and built an LLM response-quality evaluation framework that gates releases.

REAL-TIME, VISION & PLATFORM

- Built Python and Node.js WebSocket / REST services for **50,000+ concurrent users**, with event-driven inference workers on RabbitMQ and Kafka. Shipped Whisper STT (94% accuracy), Tacotron2 TTS and live captions (92%) into a video-calling product.
- Own **BlueMesh Vision**, a FastAPI service for vision-language models (PaddleOCR-VL, Qwen-VL), and a pluggable OCR layer (EasyOCR, RapidOCR, PP-OCRv4 ONNX) that makes scanned documents and charts searchable in RAG. Fine-tuned LayoutLMv3 to **91% F1** for invoice and contract extraction on 100K+ documents a month.
- Own Kubernetes delivery across **8 environments including two GPU/DGX clusters** (Kustomize, cert-manager TLS, edge routing). Also GCP Cloud Run and GitLab CI release pipelines.
- Designed **DebugTrail**, ML log analysis (Isolation Forest + LSTM) over **5M+ events/day**, cutting detection time 60% and false positives 70%. Built a streaming BERT moderation pipeline on Kafka (50 ms per message) that cut inappropriate content 85%.
- Delivered for US enterprise clients: **OfficeSuite Live** video conferencing for Windstream Enterprise, and **NCS**, a VCaaS app with **50,000+ downloads** on Play Store and App Store. Took the platform through client security reviews (encryption, RBAC, audit logging, HIPAA).

SELECTED PROJECTS (PYTHON, PUBLIC ON GITHUB)

DataRat: AI data analyst — **FastAPI · DuckDB · pandas · statsmodels · Claude / Gemini · React**

Connect 11 kinds of data source and get KPIs, anomalies (rolling z-score) and Holt-Winters forecasts. A 4-agent Planner → SQL → Analyst → Narrator pipeline answers plain-English questions, with a rules engine as fallback and a read-only SQL guard. 114 pytest tests. Live: salimkt.github.io/DataRat

small-model-agents: reliable agents on small models — **Python (stdlib) · MCP · pytest · GitHub Actions**

A reference implementation of schema-validated tool calling, tiered model routing and evidence-based escalation, with an MCP server and a mechanical eval harness whose CI gate blocks metric regressions.

3xDezine: home design and costing platform — **TypeScript · Fastify · PostgreSQL · Supabase · WebGPU · Kotlin**

2D plan to WebGPU 3D walkthrough with a live material estimate and GST-ready quotation. 300+ unit tests, 136 pgTAP database assertions, and Razorpay billing with idempotent webhooks. Live: salimkt.github.io/3xdezine

EDUCATION

B.Tech, Computer Science & Engineering
KMEA Engineering College, Aluva (APJ Abdul Kalam Technological University), Kerala · 2018 – 2022 · CGPA 7.53 / 10

CERTIFICATIONS & TRAINING

- Architecting with Google Compute Engine: Google Cloud / Coursera (6-course specialization)
- Fundamentals of Deep Learning: NVIDIA Deep Learning Institute
- Robotic Process Automation: UiPath
- Django Full-Stack Development internship: Aester India Pvt Ltd